Rice Image Classification Using Deep Learning

Transfer Learning with CNNs and Vision Transformers

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PROJECT OVERVIEW

Objective:
Classify five different rice varieties using limited image data
Problem Statement:
- Five rice varieties: Arborio, Basmati, Ipsala, Jasmine, Karacadag
- 15,000 images per class in full dataset
- Working with training samples 100 per class
- Goal: Achieve high classification accuracy

Key Approaches:
1. Transfer Learning with Pre-trained CNNs
2. Vision Transformer (ViT) Feature Extraction + SVM

DATASET DESCRIPTION

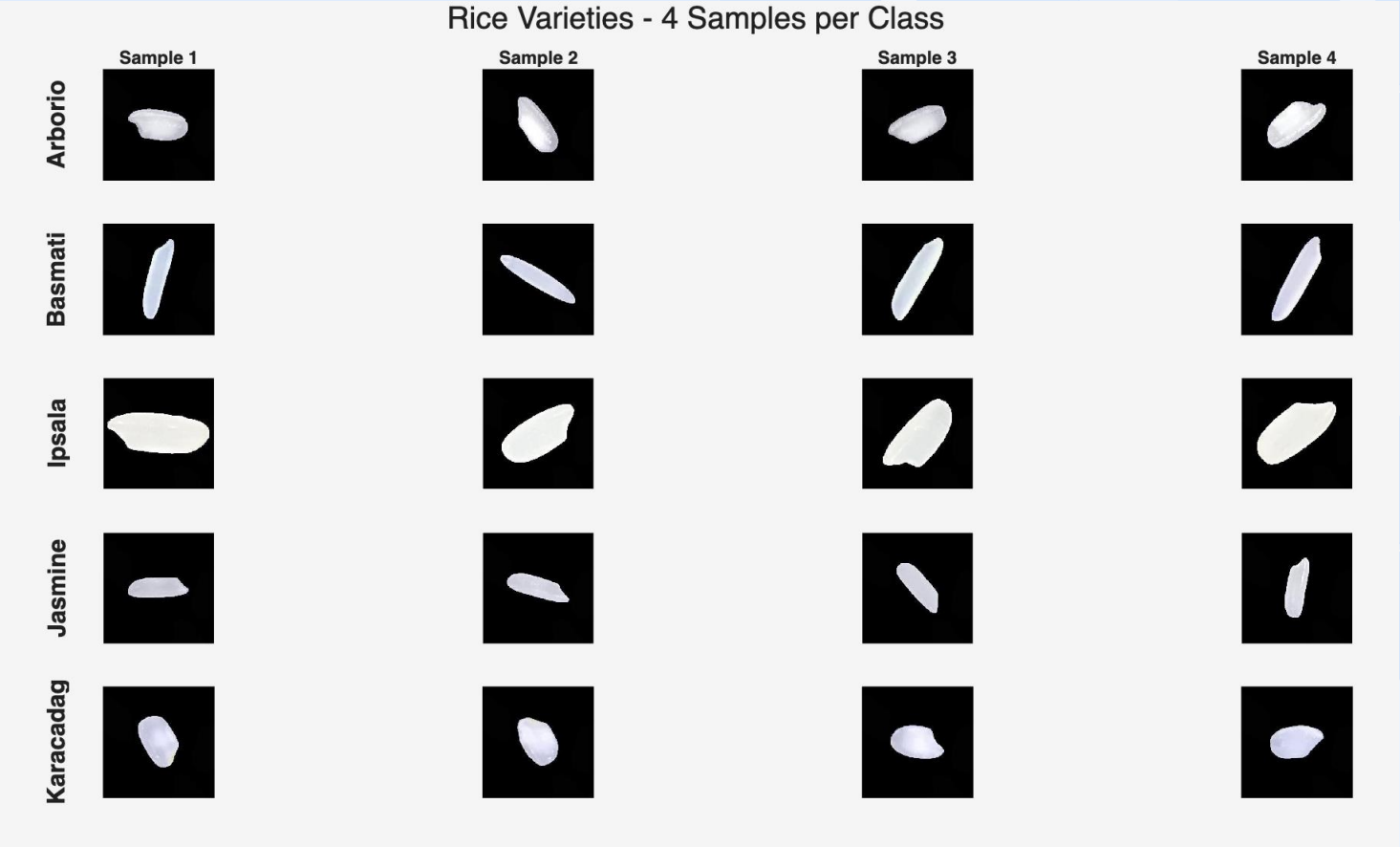
- **Classes:** 5 rice varieties
- Arborio
- Basmati
- Ipsala
- Jasmine
- Karacadag

Dataset Split:

- Training: 70% (70 images per class)
- Validation: 30% (30 images per class)
- Total samples used: 100 images per class

Data Augmentation Applied:

- Random X Reflection
- Random X/Y Translation (-30 to 30 pixels)



Model	Key Architecture Feature
AlexNet	8 layers with 3 fully connected layers
GoogleNet	Inception modules for efficient computation
ResNet50	Residual connections (50 layers deep)
VGG16	16 layers with simple, uniform architecture

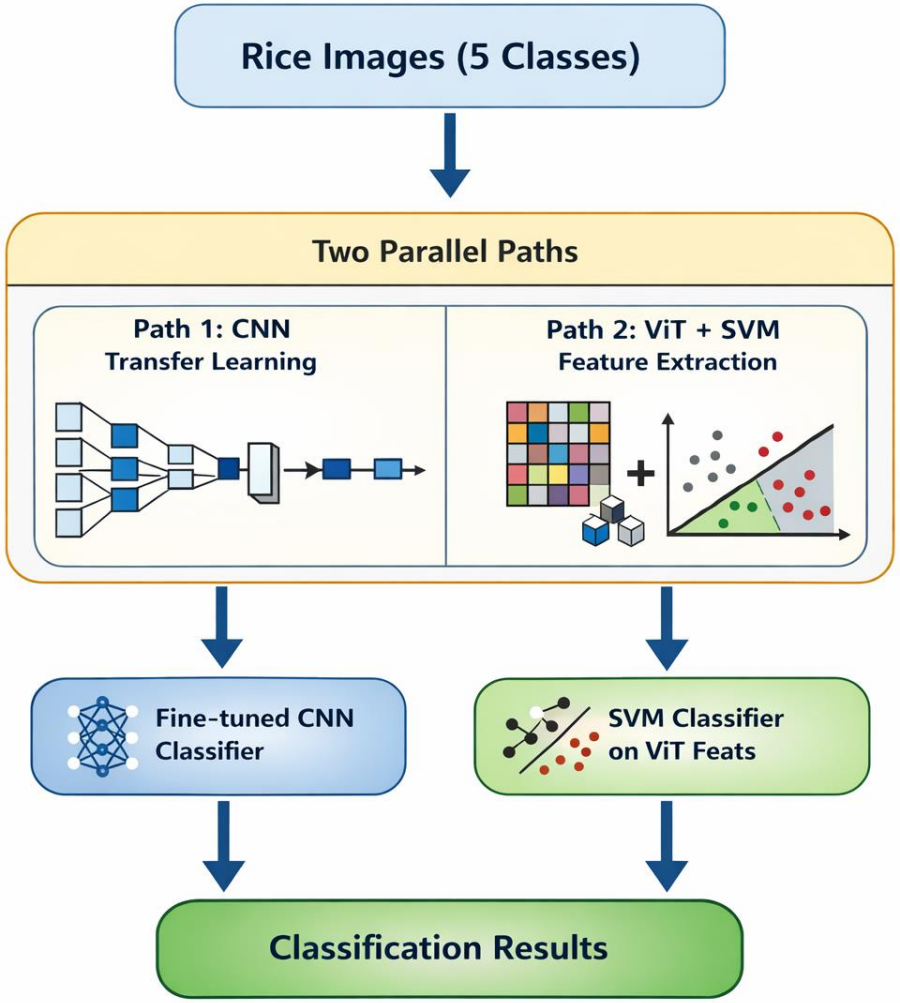
Pre-trained CNN Models Used:

Transfer Learning Modifications:

- Final classification layers removed
- New fully connected layer added (5 classes)

Training Setup:

- 100 images per class (70 train / 30 validation)
- Data augmentation: random reflections, translations
- Mini-batch size: 16
- Optimizer: SGD



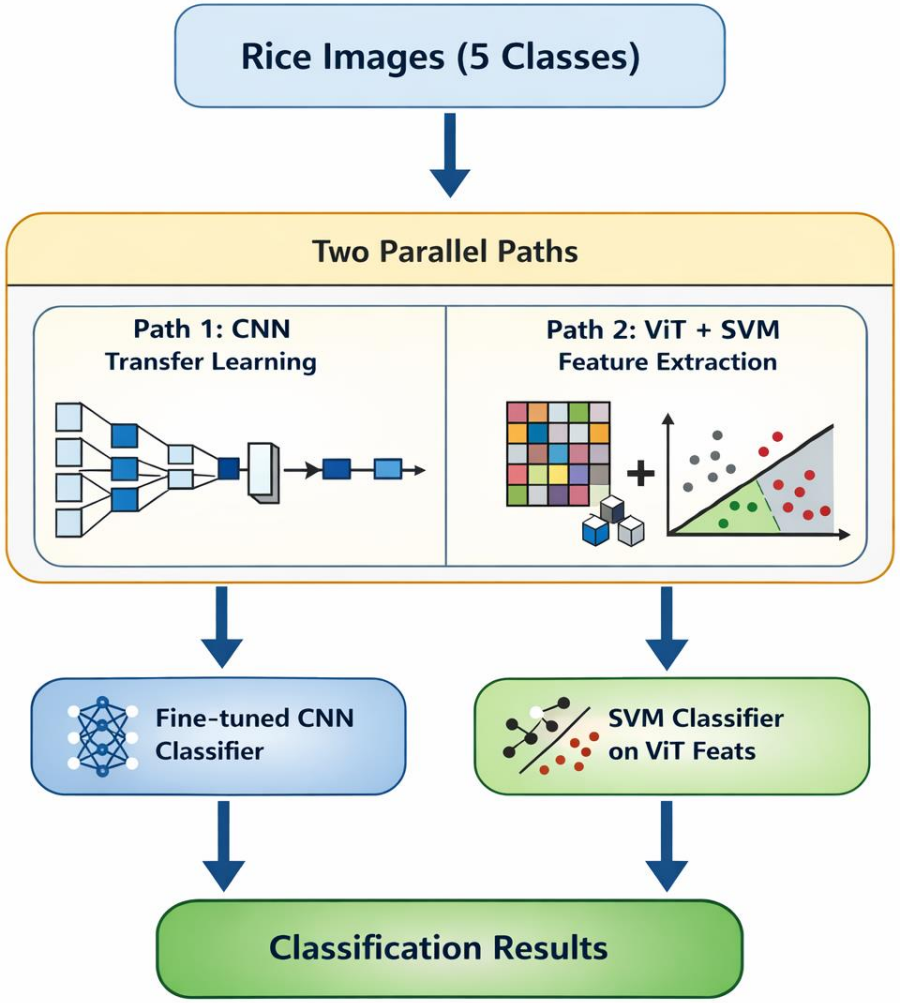
Vision Transformer + SVM Approach

Why Vision Transformer?

- Self-attention mechanism captures global image features
- Modern Approach
- Strong feature representation capabilities

Implementation Pipeline:

1. **Feature Extraction:** ViT base-16-imagenet-384 – 1000 feature extracted
2. **Feature Processing:** Normalization
3. **Classification:** Support Vector Machine

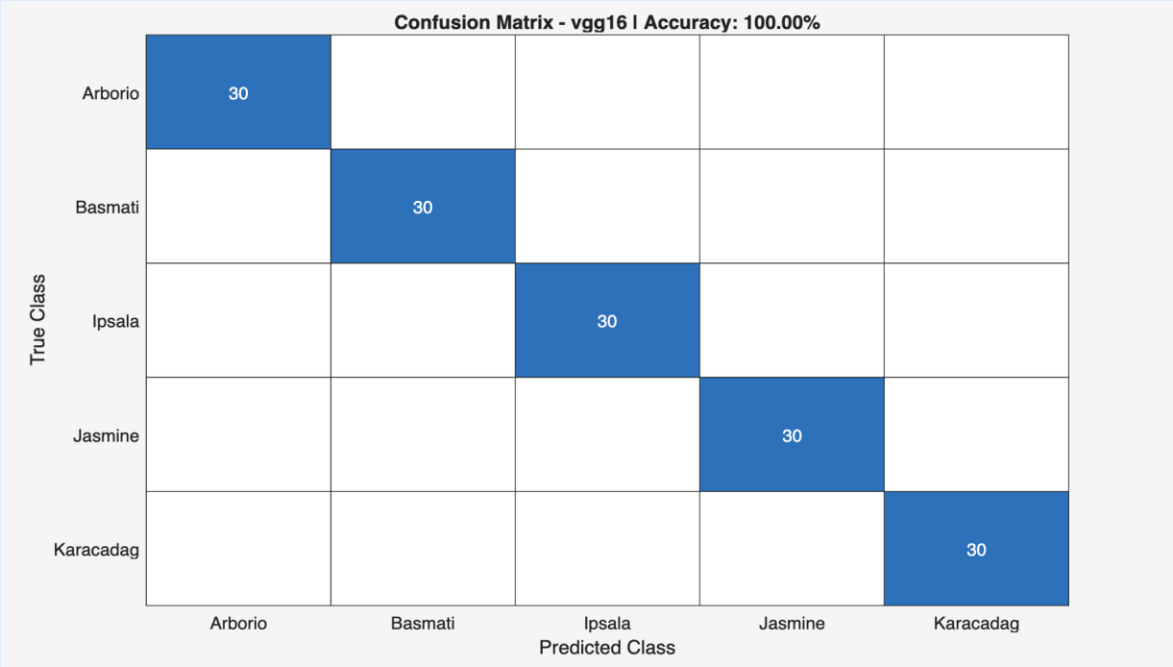


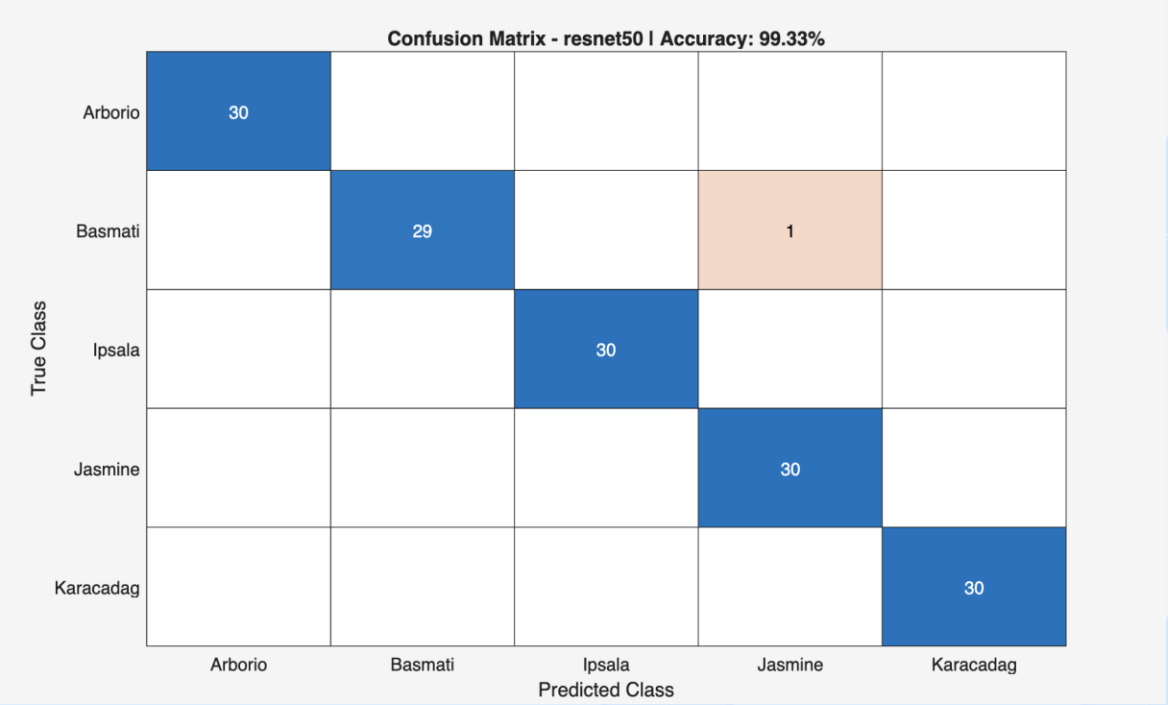
Experimental Results - CNN Models

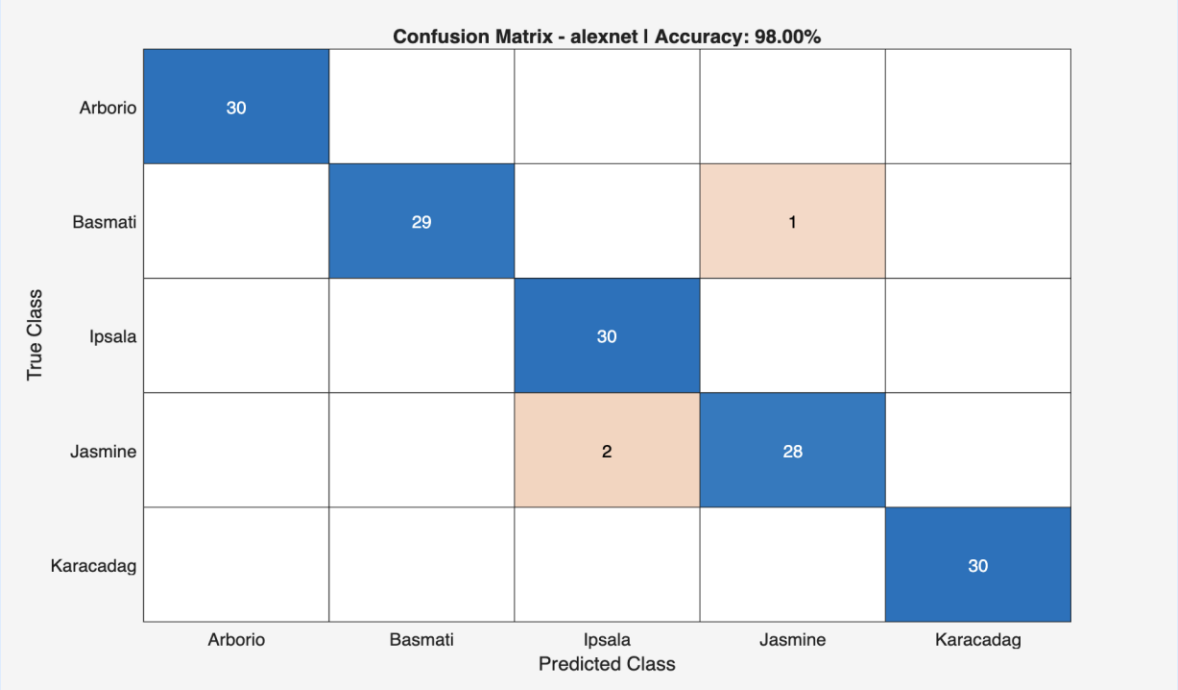
Model	Accuracy	Observations
VGG16	**100.00%**	Perfect classification
ResNet50	99.33%	Near-perfect performance
AlexNet	98.00%	Strong baseline
GoogleNet	97.33%	Good but slightly lower

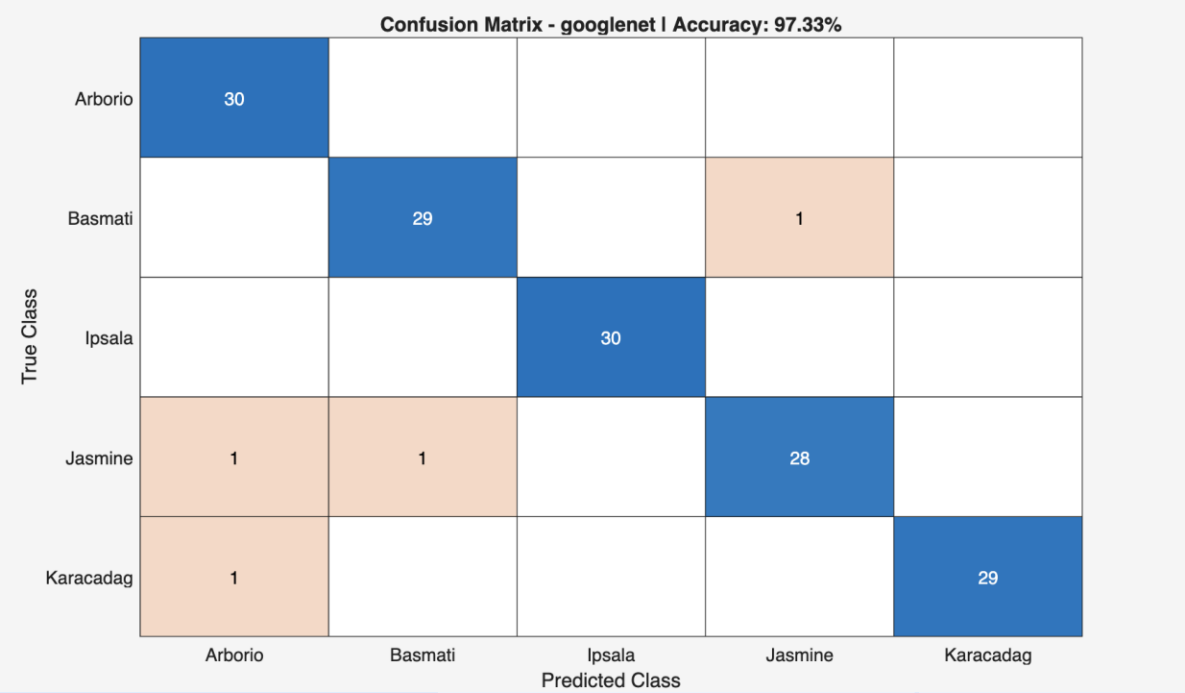
Accuracy Comparison (100 images/class):

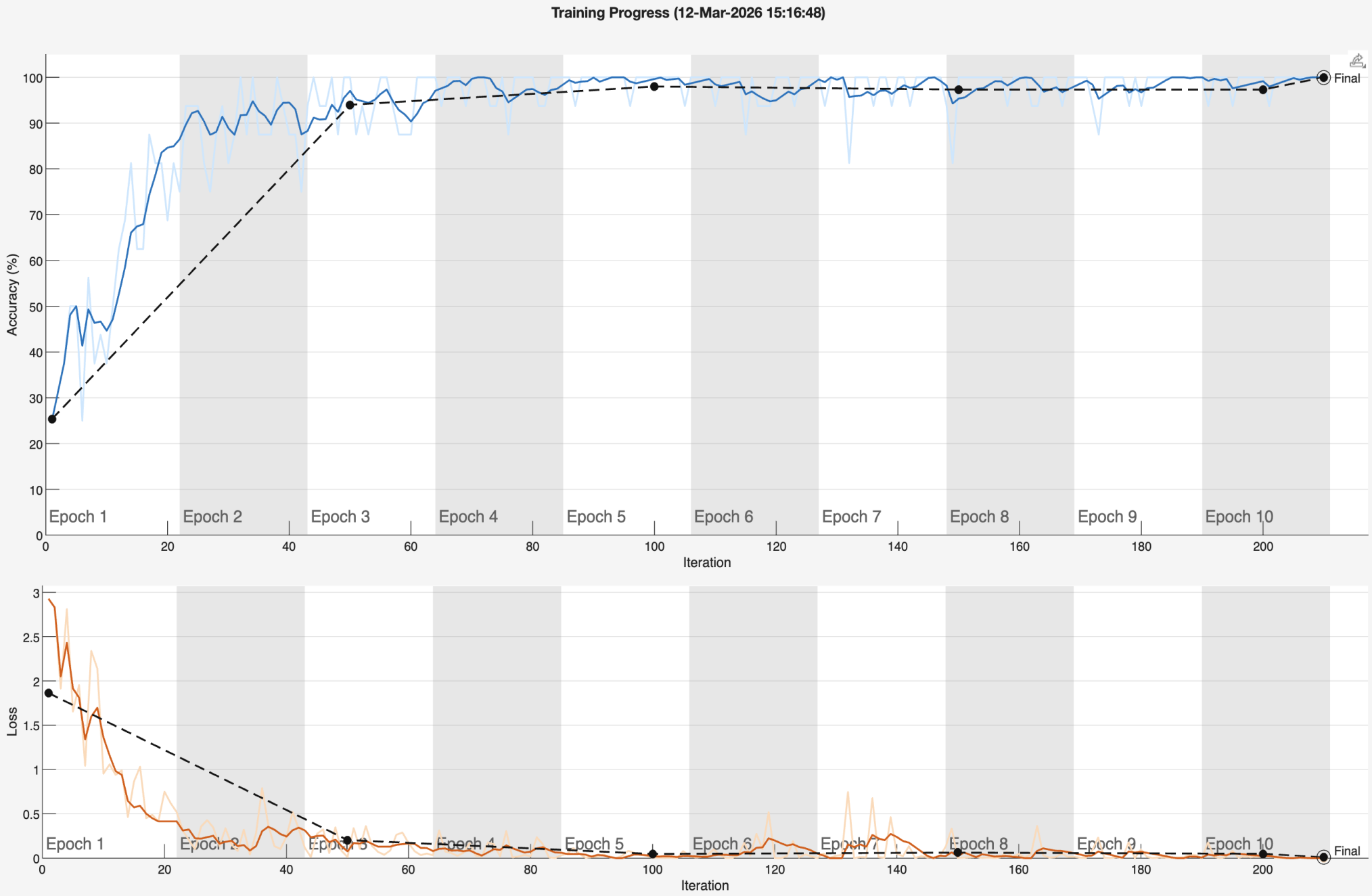
Key Insight: All CNN models achieved >97% accuracy, demonstrating the power of transfer learning.


Confusion Matrix - vgg16 | Accuracy: 100.00%
True Class
Arborio
Basmati
Ipsala
Jasmine
Karacadag
30
30
30
30
30
Arborio
Basmati
Ipsala
Jasmine
Karacadag
Predicted Class


Confusion Matrix - resnet50 | Accuracy: 99.33%
True Class
Arborio
Basmati
Ipsala
Jasmine
Karacadag
30
29
1
30
30
30
Arborio
Basmati
Ipsala
Jasmine
Karacadag
Predicted Class


Confusion Matrix - alexnet | Accuracy: 98.00%
True Class
Arborio
Basmati
Ipsala
Jasmine
Karacadag
30
29
1
30
2
28
30
Arborio
Basmati
Ipsala
Jasmine
Karacadag
Predicted Class


Confusion Matrix - googlenet | Accuracy: 97.33%
True Class
Arborio
Basmati
Ipsala
Jasmine
Karacadag
30
29
1
30
1
1
28
1
29
Arborio
Basmati
Ipsala
Jasmine
Karacadag
Predicted Class



Results

Validation accuracy:	100.00%
Training finished:	Max epochs completed

Training Time

Start time:	12-Mar-2026 15:16:48
Elapsed time:	32 min 22 sec

Training Cycle

Epoch:	10 of 10
Iteration:	210 of 210
Iterations per epoch:	21
Maximum iterations:	210

Validation

Frequency:	50 iterations

Other Information

Hardware resource:	Single CPU
Learning rate schedule:	Constant
Learning rate:	0.0001

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Accuracy
- Training (smoothed)
- Training
- Validation

Loss
- Training (smoothed)
- Training
- Validation

Vision Transformer + SVM Results

ViT + SVM Performance:

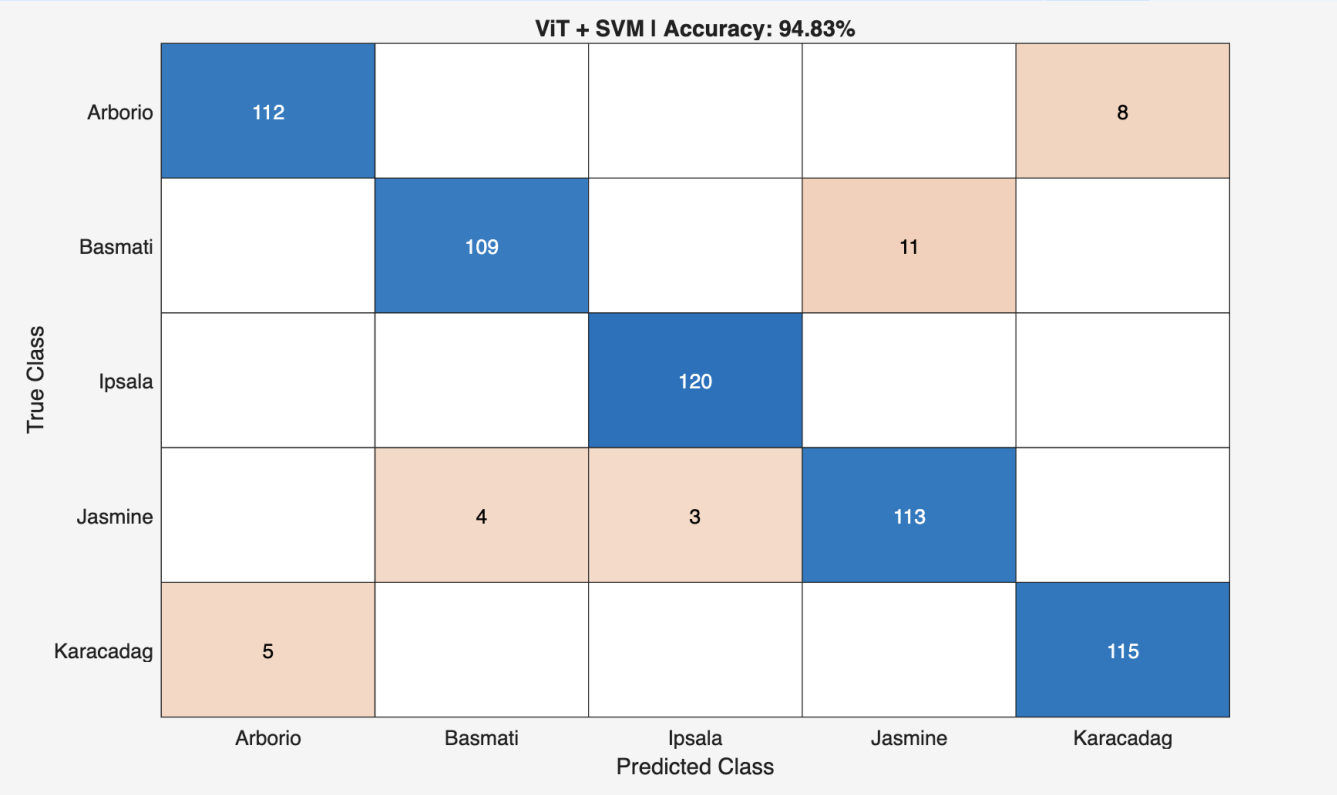
Dataset Size	Accuracy
200 images/class	94.83%

Key Observations:

- Strong performance without fine-tuning the ViT
- Feature extraction approach is efficient
- Slightly lower than top CNN models but impressive

Confusion Matrix Insight:

- Demonstrates the power of ViT features



The Vision Transformer + SVM approach demonstrated improved performance with increased training data: accuracy rose from 90.00% with 100 images per class to approximately 95% with 400 images per class, showing a clear positive correlation between dataset size and model accuracy.

Conclusions

Main Takeaways:

1. Transfer Learning is Highly Effective

1. Even with only 100 images per class, >97% accuracy achieved
2. VGG16 achieved perfect classification (100%)

2. CNN vs. ViT+SVM

1. CNNs: Superior performance on this specific task
2. ViT+SVM: Competitive alternative (94.83%) with different advantages

3. Dataset Size Impact

1. Small dataset (100 samples/class) sufficient for high accuracy
2. Data augmentation crucial for generalization

4. Practical Implications

1. Rice variety classification feasible with minimal training data
2. Multiple viable approaches available

Future Work & Recommendations

- Increase training dataset
- Hyperparameter optimization (Adam optimizer, batch size, epochs)
- Fine-tune Vision Transformer end-to-end
- Test alternative classifiers (Random Forest, ANN) on ViT features

References

1. **ResNet-ViT-SVM: A new hybrid architecture proposal**
   ScienceDirect - Experimental comparisons on date fruit
   https://www.sciencedirect.com/science/article/pii/S0889157525013353

2. **Classification of rice varieties with deep learning methods**
   ScienceDirect - Deep learning applications in agriculture

https://www.sciencedirect.com/science/article/pii/S0168169921003021